

STAT 24410/30030, Homework 5
Due Thursday, November 7, at 2:00pm

1. **(20 pts)** Lecture Notes, Chapter 4, Exercise 1.
2. **(20 pts)** Lecture Notes, Chapter 4, Exercise 2.
3. **(20 pts)** Lecture Notes, Chapter 4, Exercise 3.
4. **(20 pts)** Lecture Notes, Chapter 4, Exercise 4.

5. **(20 pts)** In class, we talked about several ways to build confidence intervals for the parameter of the exponential distribution. The setting is as follows: suppose that $T_1, \dots, T_n$ are independent and that each is distributed $\text{Exp}(\lambda)$. We denote with $\bar{T}_n$ the sample mean based on these observations and we calculated the MLE:

$$\hat{\lambda} = \frac{1}{\bar{T}_n} = \frac{n}{\sum T_i}.$$

The three confidence intervals we will explore empirically are:

- (a) Approximate the distribution on $\hat{\lambda}$ by its normal asymptotic distribution and calculate a 95% CI based on that (you need to calculate the Fisher information).
- (b) Use the pivot statistic we introduced in class, $\lambda \sum T_i$ and its exact distribution (Gamma) to calculate a 95% CI.
- (c) Use the pivot statistic above, but, instead of using the Gamma, estimate its distribution quantiles using the bootstrap - use those quantile to construct a 95% CI.

Choose $\lambda = 3$ and generate 10000 samples of 20 random variables $T_1, \dots, T_{20}$ with distribution $\text{Exp}(3)$. For each of the 10000 samples, construct the three CIs. For each CI, calculate its coverage (the proportion of simulations where the true value is in the interval), and report the 3 numbers. Show your work and include all the code you used.